

ELE459 – Lab Reports

A lab report for ELE 459 should be written as if you were documenting a project you performed at work. The people who read this document would include your supervisor and other engineers. The report might even be read several years after the work was done. Thus, the report should describe all of the details, results, and derivations needed to reproduce the work. It should define all symbols and variables used, and not assume that the reader worked on the project with you.

All reports should contain the following sections.

1. **Title Page.** This should list the title of the lab, your name, the date(s) on which you performed the lab.
2. **Summary.** The first thing that your employer wants to know when they pick up a report is, “*What is this?*” So, the first thing you should write is a brief description of the objectives or goals of the lab, which include design, simulation, and hardware implementation of a control system. The Summary section does not contain block diagrams, equations, or graphs. The Summary consists of words only. Do not use variable names in the Summary. Do not use personal pronouns (e.g. “I” or “We”) in a technical report. The passive voice is acceptable.
3. **Results.** This section should report the specific numerical results of your work, including sampling interval, settling time, choice of pole locations, and stability margins, and description of the hardware results. This is a short section (half page) and does not contain graphs or block diagrams.

4. Design and Simulation

- Describe the design of the control system, including choice of sampling interval, choice of closed-loop pole locations, and stability margins. (See Section 5 of the Lab 2 Hand-out.) **Show simulation results using Matlab graphs, not Simulink scopes.**
- Do not use personal pronouns in a technical report (e.g. “I” or “We”). The passive voice is acceptable in such reports. For example, instead of saying “*I built a Simulink model,*” say “*A Simulink model was built.*”
- Give Matlab plots for x_1 and u and explain how the settling time was chosen. Also give Matlab plots for x_1 and u in response to a tap event. **Do not use pictures of Simulink scopes; plot the data using the plot command.**
- Explain how the filter speed f was chosen.
- The captions for the Simulink diagrams should explain the functions of each of the major blocks or groups of blocks. Give a listing of the Matlab code used to design the control system.

5. Hardware Implementation

- Give details about the hardware set up. In particular, describe the issues regarding the encoder measurement of pendulum position and describe how the hardware system was initialized and run.
- Give the Simulink diagram (with explanation) that controlled the hardware.
- Describe the safety feature in the hardware Simulink model.
- Give Matlab plots for x_1 and u of the hardware response to a tap on the pendulum. Compare with the appropriate simulation results.